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NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY
SECOND SEMESTER – B.TECH.
END-SEMESTER THEORY EXAMINATION, JULY 2026

Course Code: ECECC08/ ECECC305/ ICECC305/ VTECC305/ ECECC08
Course Title: Digital Logic Systems
Time: 3:00 Hours

Max. Marks: 50

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	(i) Perform the following using 2's complement arithmetic: $101101110_2 - 110011011_2$ (ii) Simplify using Boolean algebra: $F = (A + B')(A + C)(A' + B + C')$	5	CO1
1b	(i) Minimize $F(A,B,C,D) = \Sigma m(0,2,5,7,8,10,13,15) + d(1,9)$ using K-map. (ii) Implement the minimized expression in using NAND and NOR logic gates.	5	CO1
1c	(i) Represent −91 in 8-bit Sign-Magnitude, 1's Complement, and 2's Complement forms. (ii) Add (+53) and (−91) using 8-bit 2's complement arithmetic and determine whether overflow occurs.	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Implement $F(A,B,C) = \Sigma m(1,2,5,7)$ using a 3-to-8 decoder and an 8:1 Multiplexer.	5	CO2
2b	Design a 4-bit Magnitude Comparator and derive expressions for A>B, A=B and A<B.	5	CO2
2c	(i) A decimal-to-BCD encoder receives D8 = 1. Find the BCD output. (ii) For the input sequence D2, D4, D6, D9, determine the corresponding BCD outputs.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Design a JK Flip-Flop using a D Flip-Flop by deriving the required input equation using a Karnaugh map, and draw the corresponding logic circuit.	5	CO3
3b	Explain the Master-Slave JK Flip-Flop and discuss how it eliminates race-around.	5	CO3
3c	Design a synchronous counter that follows the sequence $0 \rightarrow 1 \rightarrow 3 \rightarrow 7 \rightarrow 0$ using D flip-flops. Derive the input equations using Karnaugh maps and implement the circuit.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	Write a Verilog HDL code to implement a 1-to-8 demultiplexer using behavioural modelling. Explain its functionality.	5	CO4
4b	Write a Verilog HDL code to implement a 4-bit magnitude comparator and explain its functionality.	5	CO4

4c	Write a Verilog HDL code to implement a 4-bit ripple carry adder using structural modeling. Explain its operation.	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	Discuss the performance parameters of a digital logic family, including fan-in, fan-out, propagation delay, noise margin, power dissipation.	5	CO5
5b	Explain the operation and characteristics of the TTL logic family. Discuss its advantages and limitations.	5	CO5
5c	Explain the working principle of a Successive Approximation Register (SAR) ADC with the help of a block diagram.	5	CO5